

Supplementary Materials for  
**Structural robustness and temporal vulnerability of the starvation-responsive  
metabolic network in healthy and obese mouse liver**

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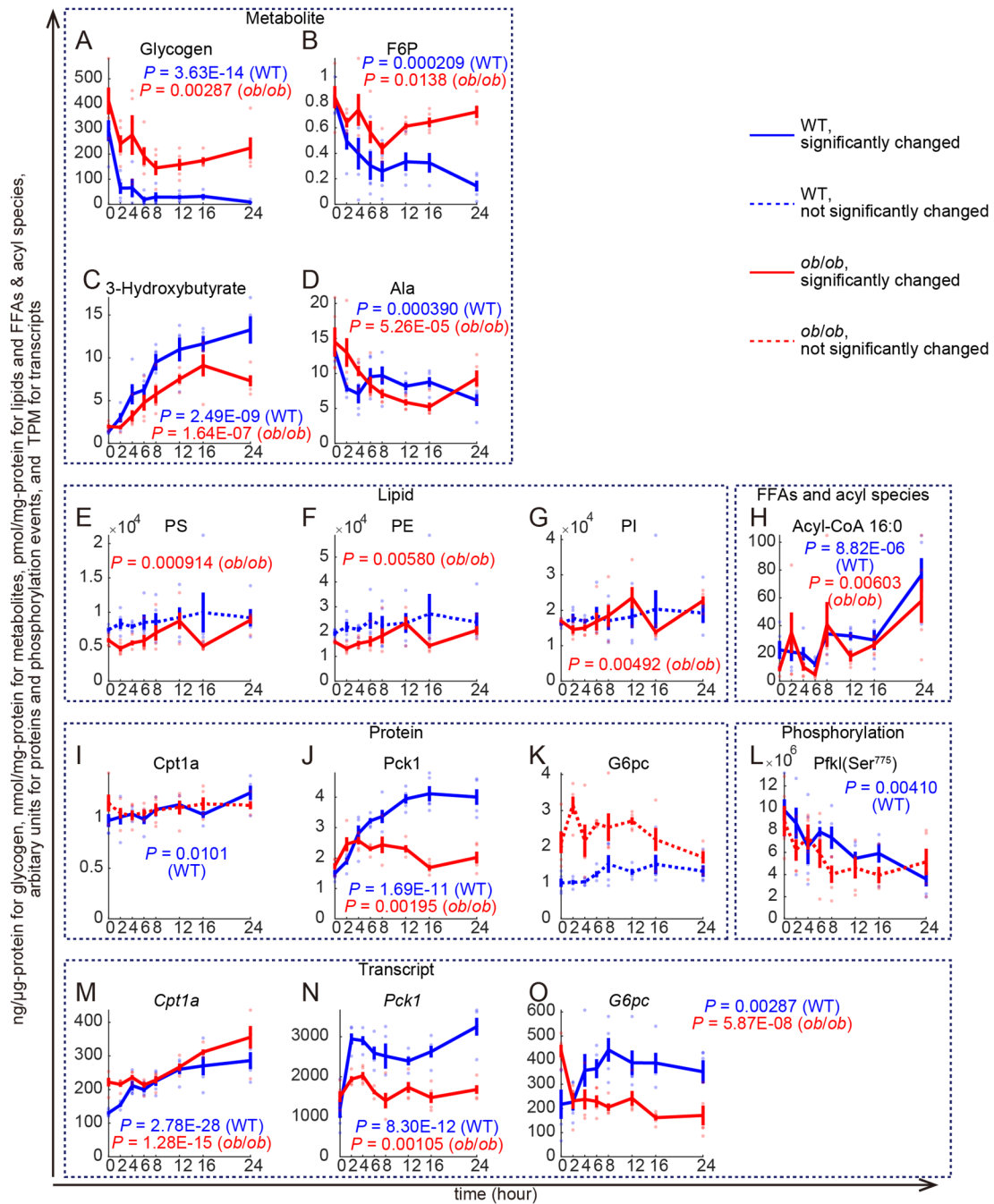
**The PDF file includes:**

Figs. S1 to S12  
Tables S1 to S8  
Legends for data files S1 to S9

**Other Supplementary Material for this manuscript includes the following:**

Data files S1 to S9  
MDAR Reproducibility Checklist

**fig. S1. Time courses of responsive molecules during starvation**



**fig. S1. Time courses of responsive molecules during starvation.**

(A to O) Time courses of glycogen (A), Fructose 6-phosphate (F6P, B), 3-hydroxybutyrate (C), Ala (D), PS (E), PE (F), PI (G), Acyl-CoA 16:0 (H), Cpt1a (I), Pck1 (J), G6pc (K), phosphorylation of Ser<sup>775</sup> in PfkI (L), Cpt1a (M), Pck1 (N), and G6pc (O). The x-axis represents the starvation time (hours) and the y-axis represents the concentration of the indicated molecule:

ng/μg-protein for glycogen, nmol/mg-protein for other metabolites (F6P, Ala, 3-hydroxybutyrate); pmol/mg-protein for lipids (PS, PE, PI) and FFAs and acyl species (Acyl-CoA 16:0); arbitrary units for proteins (Cpt1a, Pck1, G6pc) and phosphorylation (Ser<sup>775</sup> in Pfkfb3). TPM for transcripts (*Cpt1a*, *Pck1*, *G6pc*). Blue lines are the responses to starvation in WT mice, and red lines are those in *ob/ob* mice. Solid lines indicate that the responses were significant; dotted lines indicate non-significant responses. Data are shown as means and SEM. Dots represent the data from individual mice. *n* = 5 biological replicates per group. *P* values of one-way ANOVA of the molecules or ratio with significant changes are shown. All data presented in this figure are derived from data file S1.

fig. S2. Inference of regulator activities

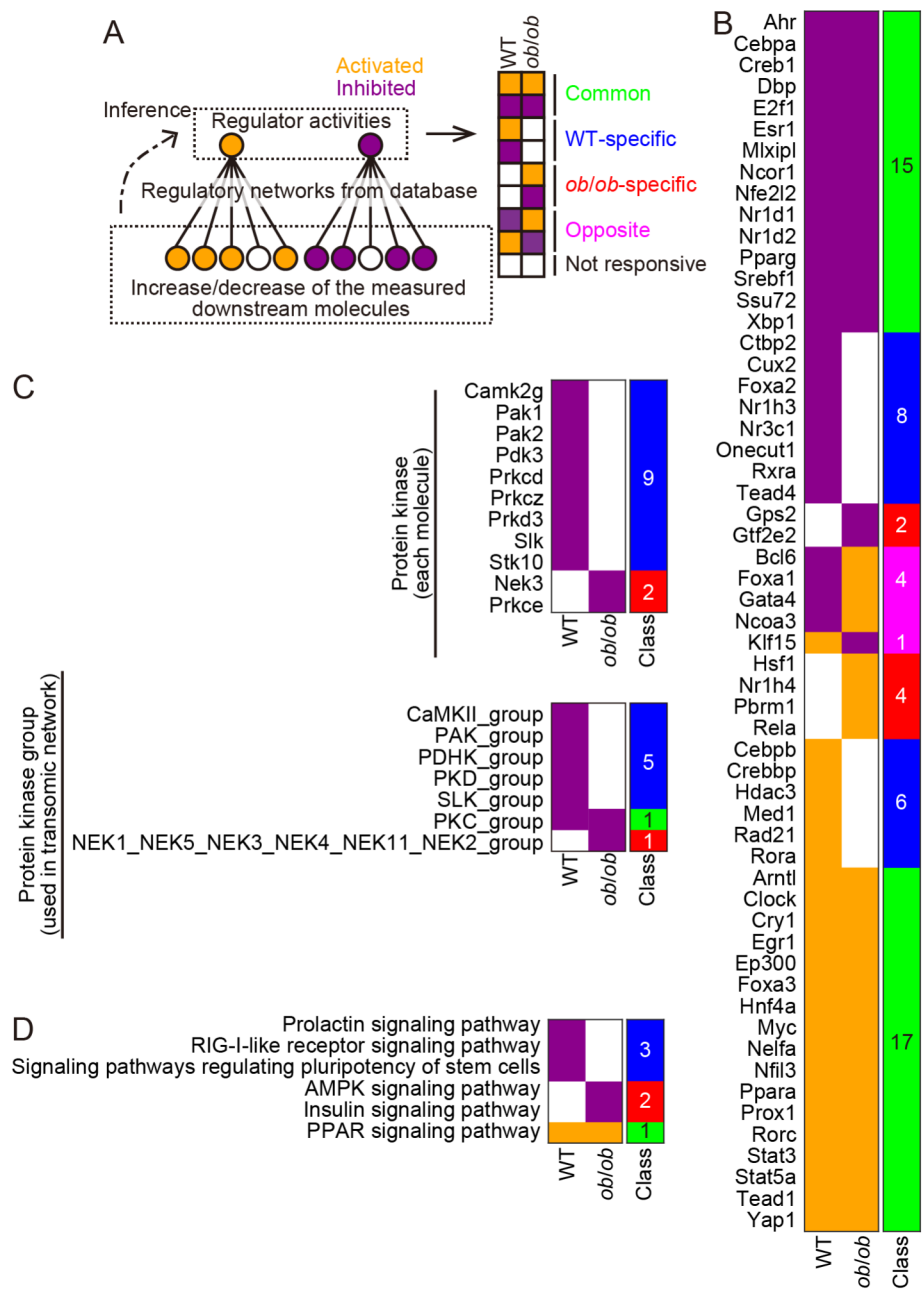
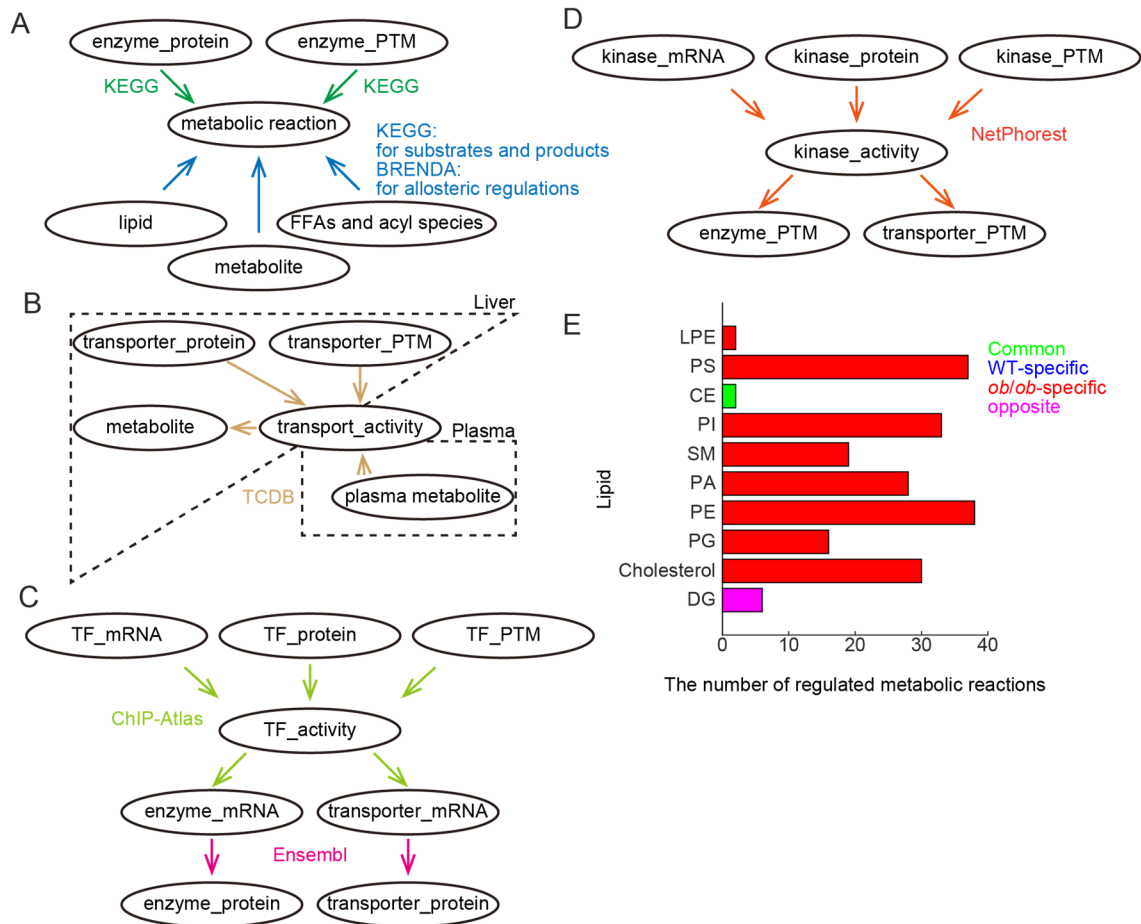


fig. S2. Inference of regulator activities.

(A) Strategy for inferring activities of upstream regulators and signaling pathways. Activated or inhibited regulators were classified into common (green), WT-specific (blue), *ob/ob*-specific (red), or opposite (magenta) groups. (B to D) Activated (orange) or inhibited (purple) transcription factors (B), protein kinases (C), or signaling pathways (D) classified into common (green), WT-specific (blue), *ob/ob*-specific (red), or opposite (magenta) groups.

**fig. S3. Regulatory relationships in the starvation-responsive trans-omic network**

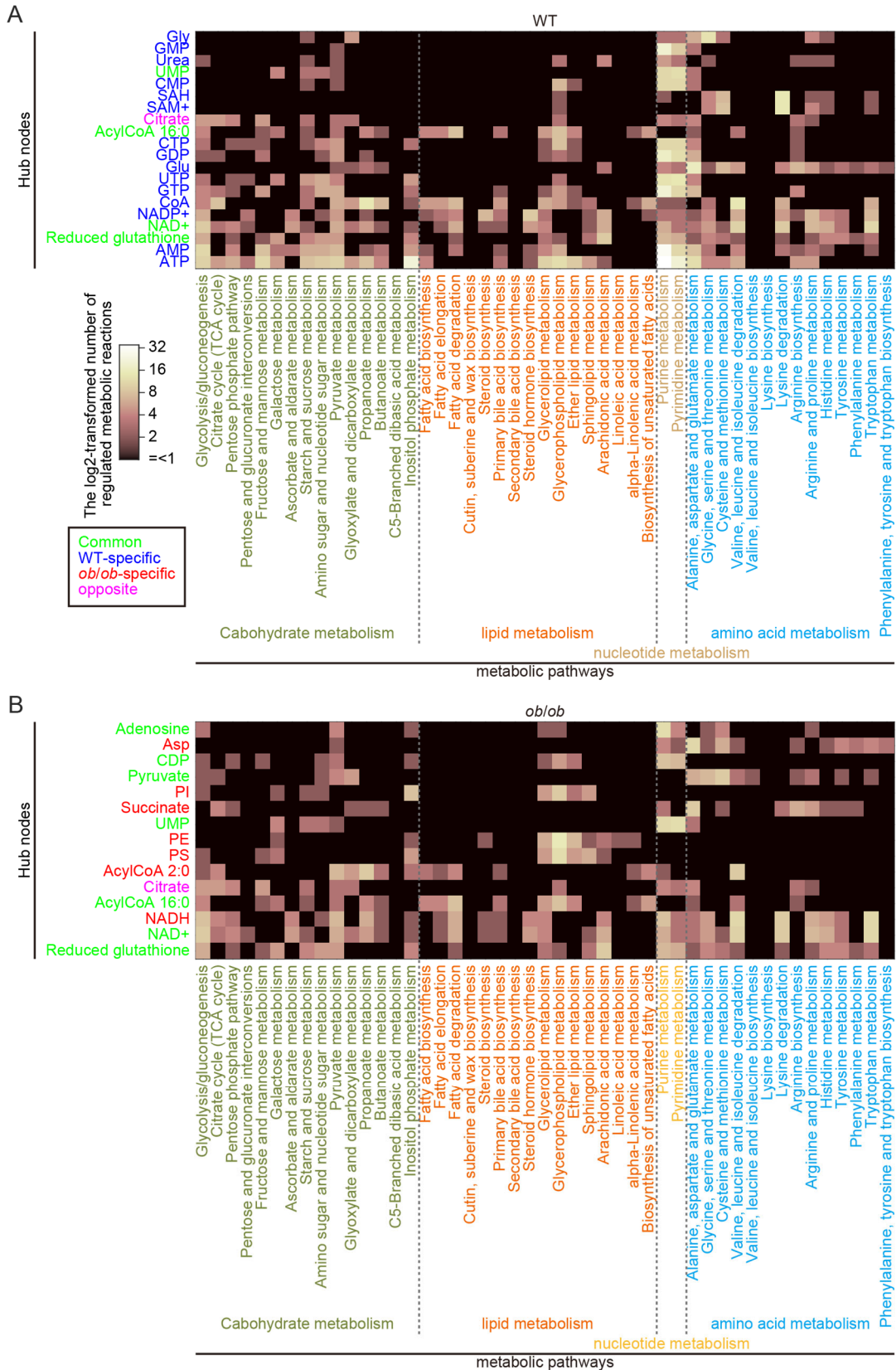


**fig. S3. Regulatory relationships in the starvation-responsive trans-omic network.**

(A) Metabolic network. The enzyme\_protein (protein abundance of metabolic enzymes) and enzyme\_PTM (phosphorylation events on metabolic enzymes) regulate metabolic reactions (identified with KEGG). Metabolites, lipids, and FFAs and acyl species regulate metabolic reactions as substrates, products (identified with KEGG), and allosteric regulators (identified with BRENDA). (B) Transport network identified with TCDB. The transporter\_protein (protein abundance of transporters), transporter\_PTM (phosphorylation events on transporters), and plasma metabolites regulate transport\_activity (activities of transport). Transport\_activity regulates metabolites in the liver, thereby regulating metabolic reactions in the liver. (C) Gene regulatory network identified with ChIP-Atlas and Ensembl. The TF\_mRNA (mRNA abundance of transcription factors), TF\_protein (protein abundance of transcription factors), and TF\_PTM (phosphorylation events on transcription factors) regulate TF\_activity (activities of transcription factors). TF\_activity regulates enzyme\_mRNA (mRNA abundance of metabolic enzymes) and transporter\_mRNA (mRNA abundance of transporters). Enzyme\_mRNA and transporter\_mRNA regulate enzyme\_protein and transporter\_protein, respectively, and were aligned with Ensembl database. (D) Phosphorylation network identified with NetPhorest. The kinase\_mRNA (mRNA abundance of kinases), kinase\_protein (protein abundance of kinase), and kinase\_PTM (phosphorylation events on kinases) regulate kinase\_activity (activities of kinases). The

kinase\_activity regulates enzyme\_PTM and transporter\_PTM. (E) The number of metabolic reactions that lipids regulate as substrates, products, and/or allosteric regulators. The colors of the bars represent responses common to both genotypes (green), *ob/ob*-specific responses (red), and responses that occur in opposite directions (magenta) to starvation.

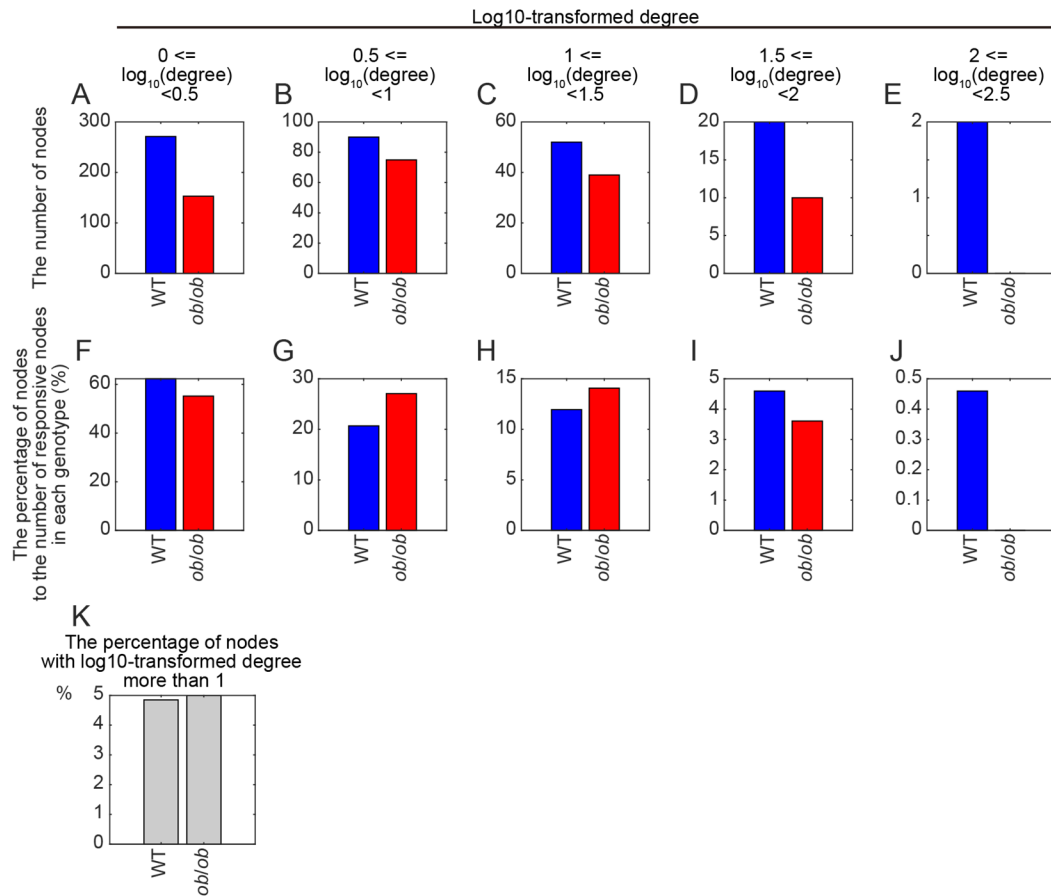
fig. S4. Metabolic pathways regulated by the hub nodes



**fig. S4. Metabolic pathways regulated by the hub nodes.**

(A and B) Heatmaps representing the number of regulated metabolic reactions in each metabolic pathway (x-axis) regulated by the hub nodes (y-axis) in the starvation-responsive metabolic network in WT (A) and *ob/ob* mice (B). Colors of the labels of the hub nodes represent the type of response: green indicates responses common to both WT and *ob/ob* mice; blue, responses specific to WT mice; red, responses specific to *ob/ob* mice; magenta, responses in WT and *ob/ob* mice that occur in opposite directions. KEGG metabolic pathways (x-axis) in carbohydrate metabolism (olive green), lipid metabolism (orange), nucleotide metabolism (brown), and amino acid metabolism (light blue) are displayed.

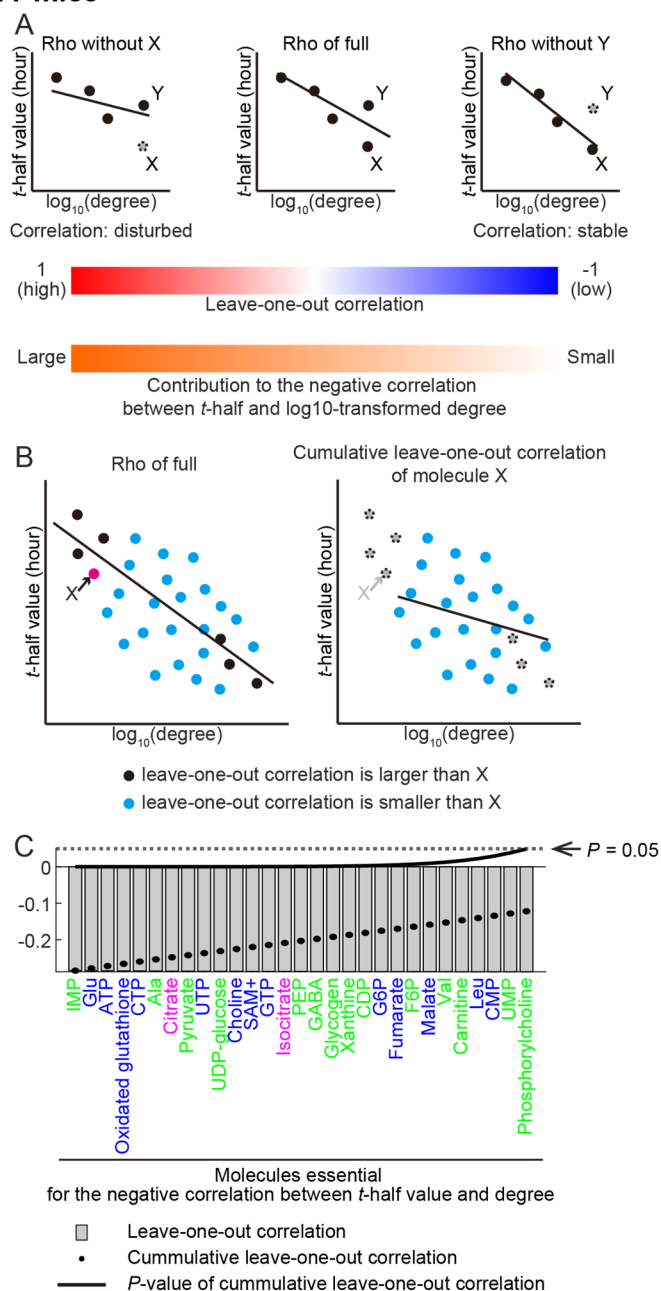
**fig. S5. Maintenance of the number of nodes with relatively large degrees in *ob/ob* mice**



**fig. S5. Maintenance of the number of nodes with relatively large degrees in *ob/ob* mice.**

(A to J) The number of nodes (A-E) and the percentage of nodes to the number of responsive nodes in each genotype (F-J). The log10-transformed degrees for the responsive nodes are indicated. (K) The percentages of nodes whose log10-transformed degree is more than 1.

**fig. S6. Identification of molecules essential for the negative correlation between degree and  $t$ -half values in WT mice**

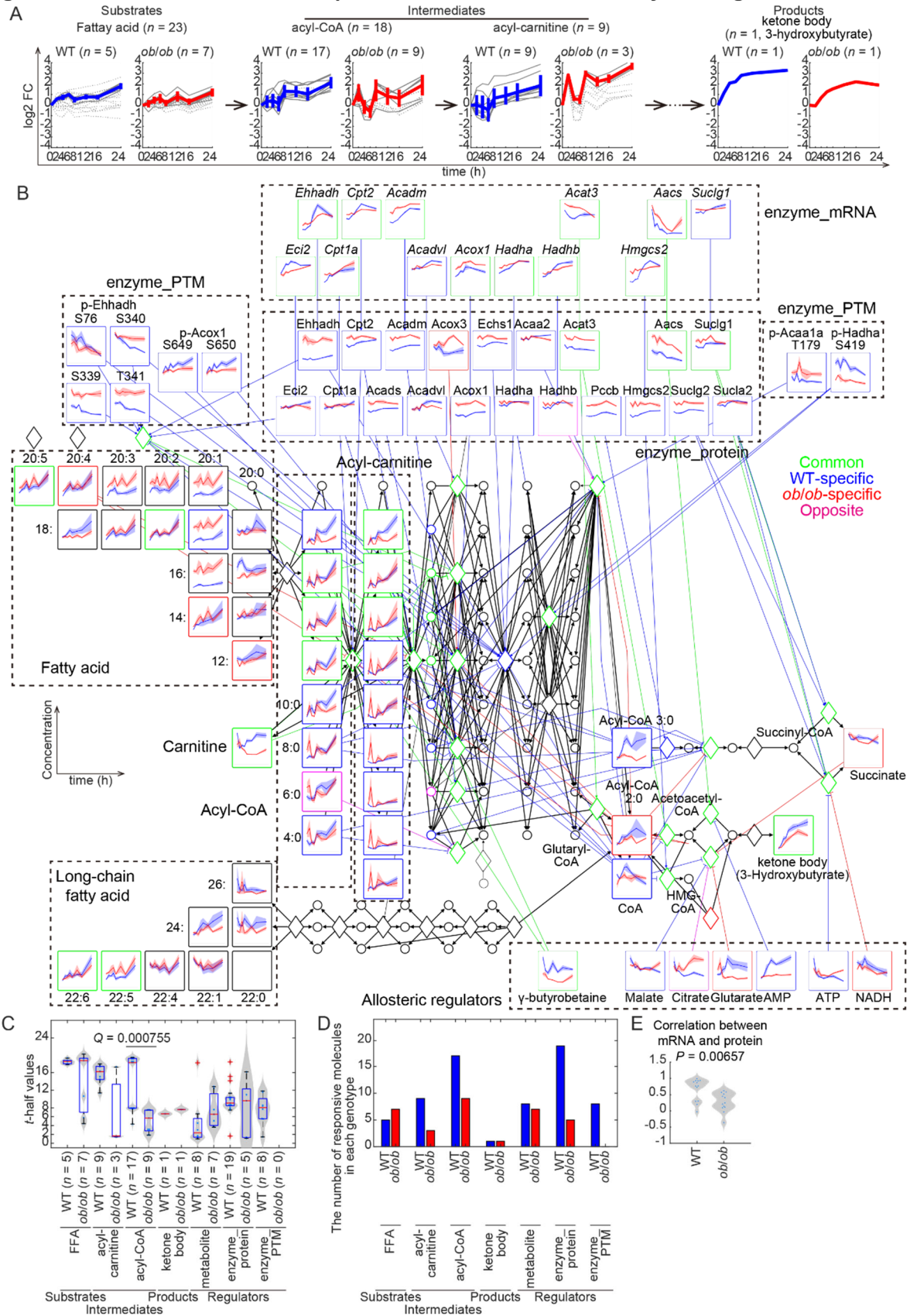


**fig. S6. Identification of molecules essential for the negative correlation between degree and  $t$ -half values in WT mice.**

(A) Schema for leave-one-out correlations. “Rho of full” represents the negative correlation between the degree and  $t$ -half value in WT mice. “Rho without X” represents the correlation coefficient between  $\log_{10}$ -transformed degree and  $t$ -half value which was calculated for all the molecules except for X, defined as the leave-one-out correlation of molecule X. The color bar graded from blue (low) to red (high) represents the leave-one-out correlation of each molecule. The orange bar represents the contribution of each molecule to the negative correlation between  $t$ -half values and  $\log_{10}$ -transformed degree. (B) Schema of the cumulative leave-one-out

correlation of molecule X (right), defined by the correlation coefficient between log10-transformed degree and  $t$ -half value that was calculated for the molecules with leave-one-out correlation smaller than X (indicated in light blue). (C) Leave-one-out correlations (gray bars), cumulative leave-one-out correlations (black dots), and  $P$ -values of the cumulative leave-one-out correlations (black solid line) of the indicated molecules. Here, we displayed molecules with  $P$  values of the cumulative leave-one-out correlations that were less than 0.05 and with leave-one-out correlations that were less than other molecules.  $n = 291$  molecules in total.

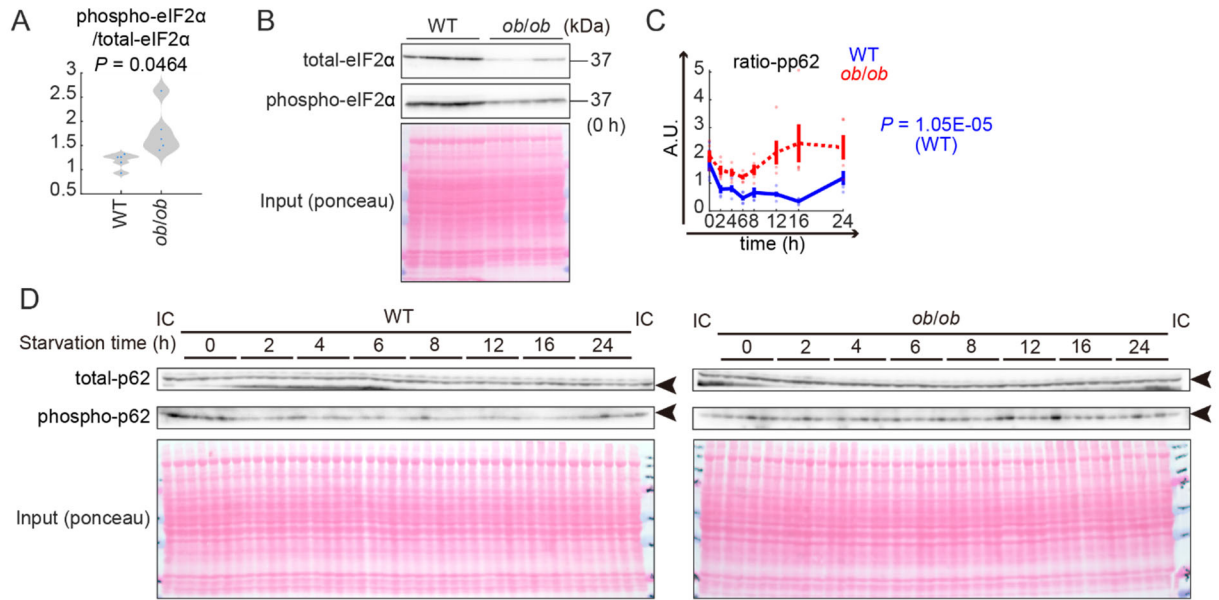
**fig. S7. The network for starvation-responsive metabolic reactions in fatty acid degradation**



**fig. S7. The network for starvation-responsive metabolic reactions in fatty acid degradation.**

(A) Time courses of fatty acids as substrates, acyl-CoAs and acyl-carnitines as intermediates, and the ketone body 3-hydroxybutyrate as a product of fatty acid degradation. Gray lines indicate the time courses of molecules in each group in each genotype. Solid lines represent significantly responsive molecules, and dashed lines represent molecules without a significant response. Bold and colored lines show means and SD of the time courses of the significantly responsive molecules. The total numbers of molecules are shown in parentheses after the names of the molecule types. The numbers of significantly changed molecules in each genotype are shown in parentheses after the genotype label. (B) The transomic network for starvation-responsive metabolic reactions in fatty acid degradation, which was manually constructed based on “Fatty acid degradation” (mmu00071), “Propanoate metabolism” (mmu00640), and “Butanoate metabolism” (mmu00650) in the KEGG database (55–57). Time courses of measured metabolites, lipids, FFAs and acyl species, enzyme\_protein, and enzyme\_PTM are shown for corresponding nodes as means and SEM.  $n = 5$  biological replicates per group. The colors of the frames represent responses common to both genotypes (green), WT-specific responses (blue), *ob/ob*-specific responses (red), and responses that occurred in opposite directions for the genotypes (magenta). The black frames indicate those were not responsive to starvation. From metabolites, lipids, and FFAs and acyl species to metabolic reactions, only allosteric regulatory connections are colored. (C) The  $t$ -half values of the indicated types of molecules in WT and *ob/ob* mice. The x-axis shows molecule types and genotypes. The y-axis shows  $t$ -half values. Light blue dots represent  $t$ -half values of each molecule. Red horizontal lines indicate median, boxes the interquartile range (25th to 75th percentile), and whiskers 1.5 times the interquartile range. Red crosses represent outliers.  $Q$  values less than 0.1 for the differences of  $t$ -half values between WT and *ob/ob* mice by Wilcoxon rank sum test followed by FDR using Benjamini-Hochberg procedure are shown. The number of molecules in each category is shown in parentheses after the genotype label. (D) The number of responsive molecules of the indicated molecule types in WT (blue) and *ob/ob* (red) mice. The x-axis shows molecule types and genotypes. The y-axis shows the number of responsive molecules. (E) The distribution of Pearson’s correlation coefficients between the time courses of enzyme\_mRNA and enzyme\_protein for each metabolic enzyme involved in fatty acid degradation for which mRNA and protein were both measured in our transcriptome and proteome data. The x-axis shows genotype and the y-axis shows Pearson’s correlation coefficient. The  $P$ -value by Wilcoxon rank sum test comparing the correlation coefficients of WT and *ob/ob* mice is shown.  $n = 13$  commonly measured molecules both in transcriptome and proteome in the transomic network in fatty acid degradation (fig. S7B). All data presented in this figure are derived from data file S1.

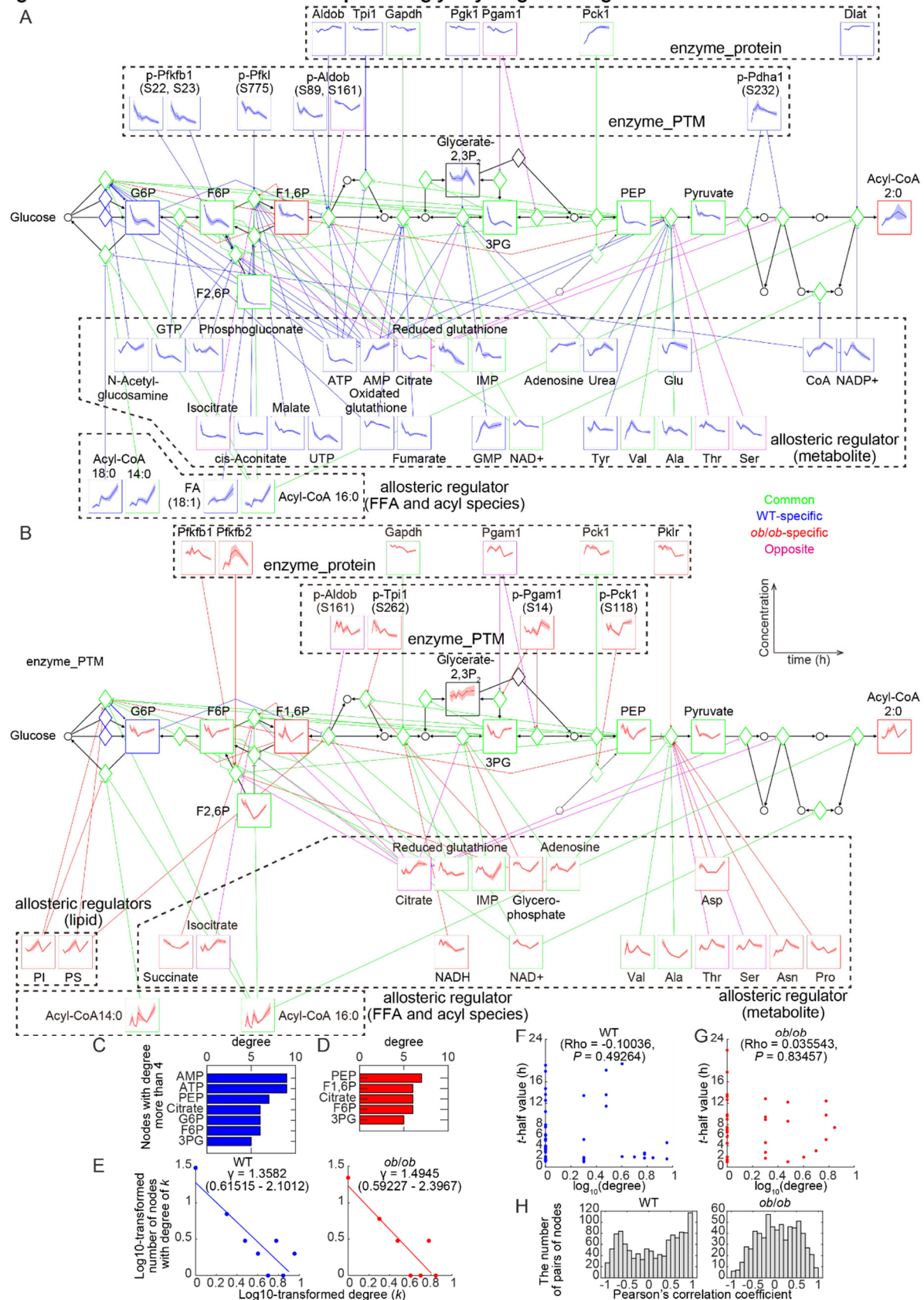
**fig. S8. Protein translation and degradation activity in *ob/ob* mice**



**fig. S8. Protein translation and degradation activity in *ob/ob* mice.**

(A) Phosphorylation ratio of eIF2 $\alpha$  in WT and *ob/ob* mice before starvation. The x-axis shows genotype and the y-axis shows the ratio of phosphorylated eIF2 $\alpha$  to total eIF2 $\alpha$ . The *P*-value by *t*-test comparing the phosphorylation ratios of eIF2 $\alpha$  between WT and *ob/ob* mice is shown. (B) Western blots quantified in (A). (C) The time course of phosphorylation ratio of p62 in WT (blue line) and *ob/ob* mice (red line). The x-axis shows starvation time (h) and the y-axis shows the ratio of phosphorylated p62 to total p62. The solid line indicates that the response was significant; the dotted line indicates a non-significant response. Data are shown as means and SEM. Dots represent the data from individual mice. *P* value by one-way ANOVA of the ratio of phosphorylated p62 to total p62 with significant changes is shown. (D) Western blots quantified in (C) at the indicated starvation time (h) for the indicated genotype. IC represents internal control to normalize signals across membranes. Arrowheads indicate the bands for total and phosphorylated p62. *n* = 5 biological replicates per group.

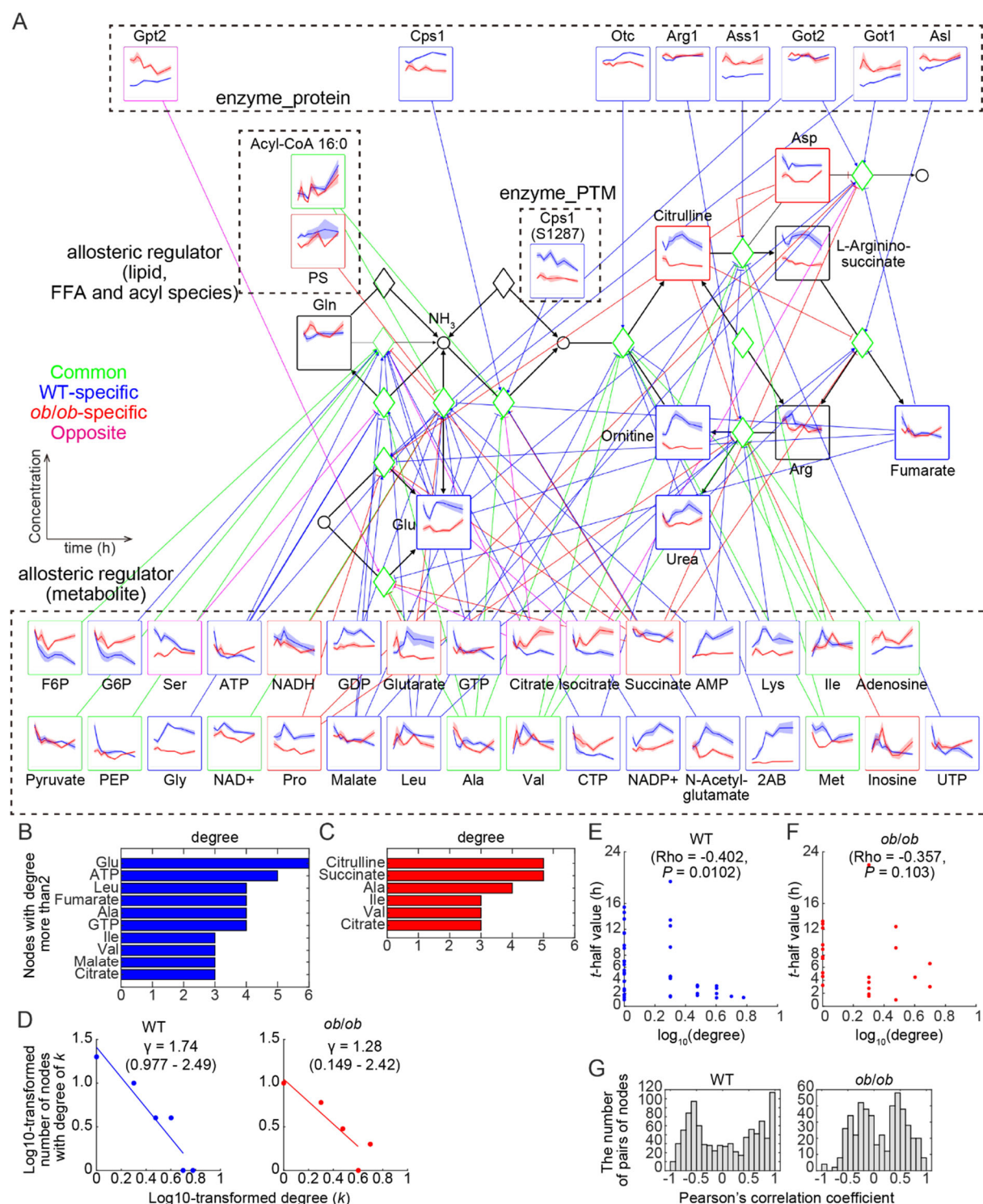
**fig. S9. The network for starvation-responsive glycolysis/gluconeogenesis metabolic reactions**



**fig. S9. The network for starvation-responsive glycolysis/gluconeogenesis metabolic reactions.**

(A and B) The starvation-responsive metabolic network for starvation-responsive metabolic reactions in glycolysis/gluconeogenesis, that were manually constructed based on “Glycolysis/gluconeogenesis” (mmu00010) in the KEGG database (55–57) in WT (A) and *ob/ob* (B) mice. Time courses of measured metabolites, lipids, FFAs and acyl species, enzyme\_protein, and enzyme\_PTM are shown for corresponding nodes as means and SEM.  $n = 5$  biological replicates per group. The colors of the frames indicate common to both genotypes (green), WT-specific responses (blue), *ob/ob*-specific responses (red), and responses that occurred in opposite directions for the genotypes (magenta). The black frames indicate those responses were not responsive to starvation. From metabolites, lipids, and FFAs and acyl species to metabolic reactions, only allosteric regulatory connections are colored. (C and D) The degree of the indicated nodes in WT (C) and in *ob/ob* (D) mice. Nodes with a degree more than 4 are shown. The x-axis represents degree (the number of edges) and the y-axis represents nodes. It should be noted that the degrees shown in (C) and (D) are not necessarily identical to those shown in (A) and (B), because if a molecule regulates a metabolic reaction both as a substrate or product, and as an allosteric regulator, the edges were counted as a single edge in (C) and (D), but were shown separately in (A) and (B). (E) Degree distributions of the nodes in the starvation-responsive metabolic network of glycolysis/gluconeogenesis in WT (blue) and *ob/ob* (red) mice. The x-axis shows the log10-transformed degree ( $k$ ); the y-axis shows the log10-transformed number of nodes with degree of  $k$ . Lines show the linear regression of the degree distributions. The slope of the lines ( $\gamma$ ), which is defined as a scaling parameter, and the 95% credible interval are shown.  $n = 49$  molecules for WT mice and  $n = 37$  molecules for *ob/ob* mice. (F and G) Scatter plots for the log10-transformed degree (x-axis) and  $t$ -half value (y-axis) in WT (blue; F) and *ob/ob* (red; G) mice. Dots represent the molecules (including phosphorylation events) in the starvation-responsive metabolic network of glycolysis/gluconeogenesis. Blue dots represent WT and red dots represent *ob/ob* mice. Spearman’s correlation coefficients (Rho) and  $P$  values are shown.  $n = 49$  molecules for WT mice and  $n = 37$  molecules for *ob/ob* mice. (H) Distribution of the pairwise Pearson’s correlation coefficients among time courses of responsive molecules in the starvation-responsive glycolysis/gluconeogenesis metabolic network in WT and *ob/ob* mice. The x-axis shows correlation coefficients; the y-axis shows the number of pairs of molecules examined for correlation.  $n = 49$  molecules for WT mice and  $n = 37$  molecules for *ob/ob* mice. All data presented in this figure are derived from data file S1.

**fig. S10. The network for starvation-responsive urea cycle metabolic reactions**

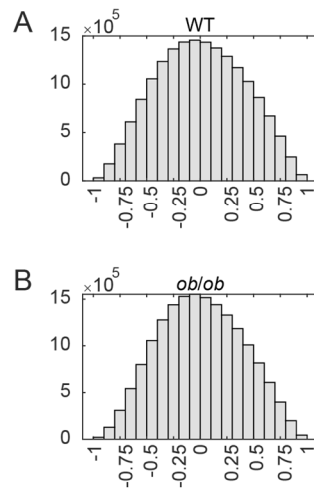


**fig. S10. The network for starvation-responsive urea cycle metabolic reactions.**

(A) The network for starvation-responsive urea cycle metabolic reactions which were manually constructed based on “Arginine biosynthesis” (mmu00220) in the KEGG database (55–57). Time

courses of measured metabolites, lipids, FFAs and acyl species, enzyme\_protein, and enzyme\_PTM are shown for corresponding nodes as means and SEM. The colors of the frames indicate common to both genotypes (green), WT-specific responses (blue), *ob/ob*-specific responses (red), and responses that occurred in opposite directions for the genotypes (magenta). The black frames indicate non-responsive reactions. From metabolites, lipids, and FFAs and acyl species to metabolic reactions, only allosteric regulatory connections are colored. **(B and C)** The degree of the indicated nodes in WT (B) and in *ob/ob* (C) mice. Nodes with degree more than 2 are shown. The x-axis represents degree (the number of edges) and the y-axis represents nodes. It should be noted that the degrees shown in (B) and (C) are not necessarily identical to those shown in (A), because if a molecule regulates a metabolic reaction both as a substrate or product, and as an allosteric regulator, the edges were counted as a single edge in (B) and (C), but were shown separately in (A). **(D)** Degree distributions of the nodes in the starvation-responsive urea cycle metabolic network in WT (blue) and *ob/ob* (red) mice. The x-axis shows the log10-transformed degree ( $k$ ); the y-axis shows the log10-transformed number of nodes with degree of  $k$ . Lines show the linear regression of the degree distributions. The slope of the lines ( $\gamma$ ), which is defined as a scaling parameter, and the 95% credible interval are shown.  $n = 40$  molecules for WT mice and  $n = 22$  molecules for *ob/ob* mice. **(E and F)** Scatter plots for the log10-transformed degree (x-axis) and  $t$ -half (y-axis) in WT (blue; E) and *ob/ob* (red; F) mice. Dots represent the molecules (including phosphorylation events) of the starvation-responsive urea cycle metabolic network. Spearman's correlation coefficients (Rho) and  $P$  values are shown.  $n = 40$  molecules for WT mice and  $n = 22$  molecules for *ob/ob* mice. **(G)** Distribution of the pairwise Pearson's correlation coefficients among time courses of responsive molecules in the starvation-responsive metabolic network of urea cycle in WT and *ob/ob* mice. The x-axis shows correlation coefficients; the y-axis shows the number of pairs of molecules examined for correlation.  $n = 40$  molecules for WT mice and  $n = 22$  molecules for *ob/ob* mice. All data presented in this figure are derived from data file S1.

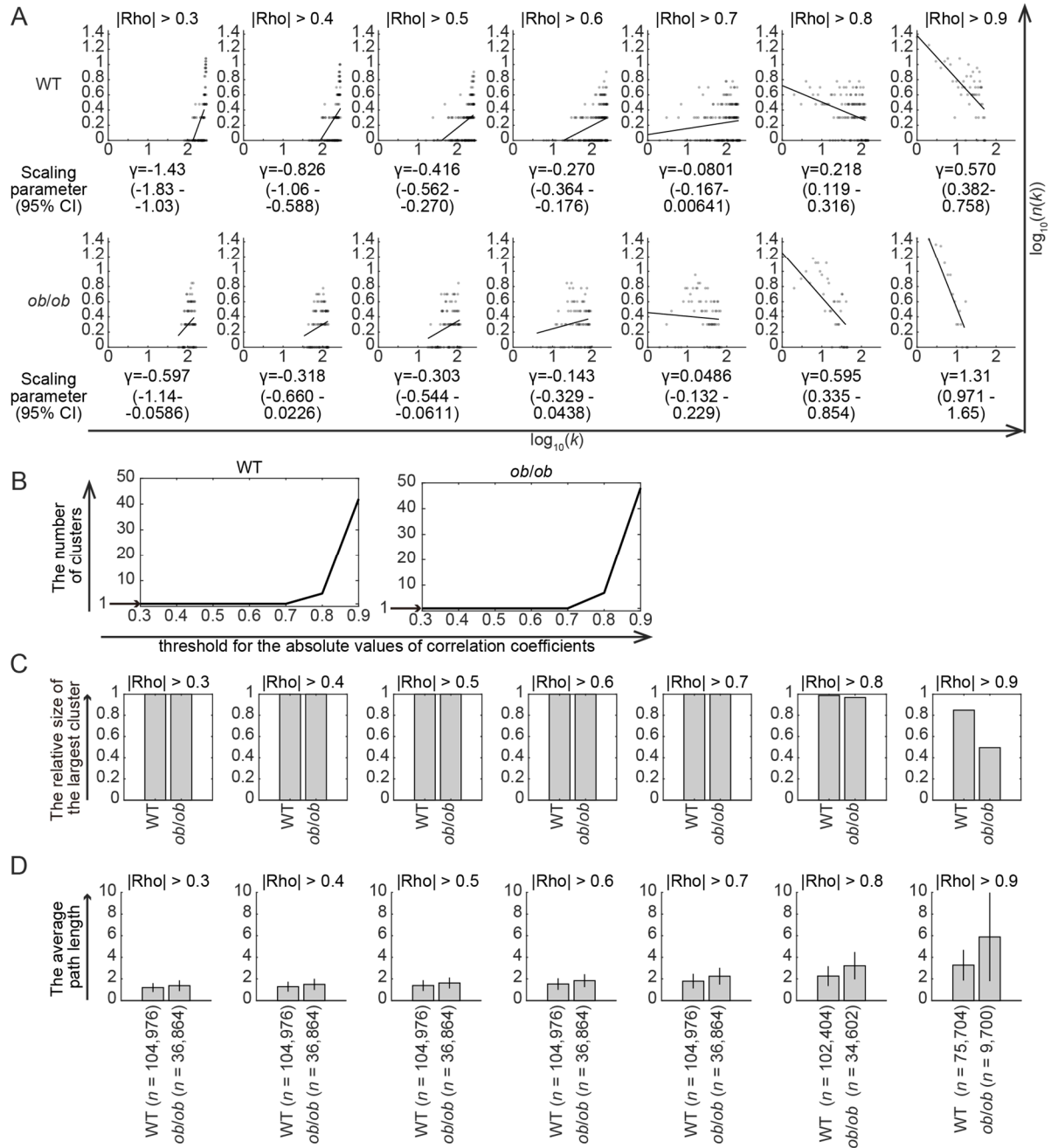
**fig. S11. Distribution of the pairwise Pearson's correlation coefficients among time courses of all the measured metabolites, lipids, FFA and acyl species, enzyme\_protein, and enzyme\_PTM**



**fig. S11. Distribution of the pairwise Pearson's correlation coefficients among time courses of all the measured metabolites, lipids, FFA and acyl species, enzyme\_protein, and enzyme\_PTM.**

(A and B) The x-axis shows the pairwise Pearson's correlation coefficients among time courses of all the measured metabolites, lipids, FFAs and acyl species, enzyme\_protein and enzyme\_PTM; the y-axis represents the number of pairs of molecules examined for correlation. The data of WT (A) and *ob/ob* (B) mice are shown.  $n = 5,866$  molecules in each genotype.

**fig. S12. The correlation metabolic network during starvation**



**fig. S12. The correlation metabolic network during starvation.**

(A to D) Scatter plots (A) showing the degree distributions of correlation metabolic network corresponding to each threshold of absolute values of correlation coefficients ( $|Rho|$ ) ranging from 0.3 to 0.9. The x-axis shows  $\log_{10}(k)$ , the log10-transformed degree; the y-axis shows  $\log_{10}(n(k))$ , the log10-transformed number of nodes with degree of  $k$ . Dots denote the degree distribution of the indicated networks and lines show the linear-regression of the degree distribution. The scaling parameters ( $\gamma$ ) and the 95% credible interval are shown. Data for WT and *ob/ob* mice are shown. The number of clusters (B), relative size of the largest cluster (C) and

the average path length (D) for each threshold are also shown. For average path length, the numbers of the investigated paths are shown in parentheses after the genotype label. The error bars show standard deviation.

| Omics type            | Maximum $Q$ -value |
|-----------------------|--------------------|
| Metabolome (Liver)    | 0.0029             |
| Transcriptome         | 0.0489             |
| Proteome              | 0.0661             |
| Phosphoproteome       | 0.1645             |
| Lipidome              | 0.0502             |
| FFAs and acyl species | 0.053              |
| Metabolome (Plasma)   | 0.0016             |

**Table S1. Maximum  $Q$  values for the omic layers of starvation-responsive molecules.**  
Maximum  $Q$  values for each omic layer for molecules with  $P$  values less than 0.025.

| Common name                  | Abbreviation |
|------------------------------|--------------|
| Lysophosphatidylcholine      | LPC          |
| Lysophosphatidylethanolamine | LPE          |
| Cholesterol                  | Chol         |
| Phosphatidylethanolamine     | PE           |
| Triacylglycerol              | TG           |
| Ceramide                     | Cer          |
| Diacylglycerol               | DG           |
| Monoacylglycerol             | MG           |
| Phosphatidic acid            | PA           |
| Phosphatidylcholine          | PC           |
| Phosphatidylglycerol         | PG           |
| Phosphatidylinositol         | PI           |
| Phosphatidylserine           | PS           |
| Sphingomyelin                | SM           |
| Cholesterol ester            | CE           |
| Hexose ceramide              | HexCer       |

**Table S2. Lipid classes measured.**

|           | WT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>ob/ob</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Increased | Fatty acid degradation, Lysine degradation, Glycerolipid metabolism, Biosynthesis of unsaturated fatty acids, PPAR signaling pathway, Peroxisome, Fat digestion and absorption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PPAR signaling pathway, Fat digestion and absorption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Decreased | Pentose and glucuronate interconversions, Ascorbate and aldarate metabolism, Steroid biosynthesis, Steroid hormone biosynthesis, Tyrosine metabolism, Selenocompound metabolism, Glutathione metabolism, Thiamine metabolism, Retinol metabolism, Porphyrin metabolism, Terpenoid backbone biosynthesis, Sulfur metabolism, Metabolism of xenobiotics by cytochrome P450, Drug metabolism - cytochrome P450, Drug metabolism - other enzymes, ABC transporters, Protein export, Mismatch repair, Protein processing in endoplasmic reticulum, Necroptosis, Cellular senescence, Signaling pathways regulating pluripotency of stem cells, RIG-I-like receptor signaling pathway, Serotonergic synapse, Prolactin signaling pathway, Bile secretion | Glycolysis / Gluconeogenesis, Pentose phosphate pathway, Fructose and mannose metabolism, Fatty acid biosynthesis, Steroid biosynthesis, Steroid hormone biosynthesis, Glycine, serine and threonine metabolism, Tryptophan metabolism, Selenocompound metabolism, Glutathione metabolism, Starch and sucrose metabolism, Amino sugar and nucleotide sugar metabolism, Pyruvate metabolism, Glyoxylate and dicarboxylate metabolism, One carbon pool by folate, Nicotinate and nicotinamide metabolism, Retinol metabolism, Porphyrin metabolism, Terpenoid backbone biosynthesis, Sulfur metabolism, Aminoacyl-tRNA biosynthesis, Metabolism of xenobiotics by cytochrome P450, Drug metabolism - cytochrome P450, Drug metabolism - other enzymes, Biosynthesis of unsaturated fatty acids, ABC transporters, DNA replication, Protein export, Base excision repair, Mismatch repair, Non-homologous end-joining, Protein processing in endoplasmic reticulum, Peroxisome, AMPK signaling pathway, Longevity regulating pathway, Thermogenesis, Phototransduction, Insulin signaling pathway |

**Table S3. KEGG pathway enrichment analysis of starvation-responsive transcripts.** KEGG pathways with  $Q$ -value  $< 0.1$  are shown.

|           | WT                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>ob/ob</i>       |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Increased | Citrate cycle (TCA cycle), Fatty acid degradation, Arginine biosynthesis, Valine, leucine and isoleucine degradation, Phenylalanine metabolism, Phenylalanine, tyrosine and tryptophan biosynthesis, beta-Alanine metabolism, Propanoate metabolism, PPAR signaling pathway                                                                                                                                                      | Ribosome, Lysosome |
| Decreased | Pentose and glucuronate interconversions, Ascorbate and aldarate metabolism, Steroid hormone biosynthesis, Glutathione metabolism, One carbon pool by folate, Retinol metabolism, Porphyrin metabolism, Metabolism of xenobiotics by cytochrome P450, Drug metabolism - cytochrome P450, Drug metabolism - other enzymes, Protein processing in endoplasmic reticulum, Antigen processing and presentation, Serotonergic synapse | No enrichment      |

**Table S4. KEGG pathway enrichment analysis of starvation-responsive proteins.** KEGG pathways with  $Q$ -value  $< 0.1$  are shown.

|      |              |                                                                                                                  |
|------|--------------|------------------------------------------------------------------------------------------------------------------|
| WT   | <i>ob/ob</i> | Transcription factors                                                                                            |
| Up   | Up           | Arntl, Clock, Cry1, Egr1, Ep300, Foxa3, Hnf4a, Myc, Nelfa, Nfil3, Ppara, Prox1, Rorc, Stat3, Stat5a, Tead1, Yap1 |
| Up   | NE           | Cebpb, Crebbp, Hdac3, Med1, Rad21, Rora                                                                          |
| Up   | Down         | Klf15                                                                                                            |
| NE   | Up           | Hsf1, Nr1h4, Pbrm1, Rela                                                                                         |
| NE   | Down         | Gps2, Gtf2e2                                                                                                     |
| Down | Up           | Bcl6, Foxa1, Gata4, Ncoa3                                                                                        |
| Down | NE           | Ctbp2, Cux2, Foxa2, Nr1h3, Nr3c1, Onecut1, Rxra, Tead4                                                           |
| Down | Down         | Ahr, Cebpa, Creb1, Dbp, E2f1, Esr1, Mlxipl, Ncor1, Nfe2l2, Nr1d1, Nr1d2, Pparg, Srebf1, Ssu72, Xbp1              |

**Table S5. Activated or inhibited transcription factors during starvation.** Activated ('Up') or inhibited ('Down') transcription factors during starvation. 'NE' indicates no enrichment.

|      |              |                                                           |
|------|--------------|-----------------------------------------------------------|
| WT   | <i>ob/ob</i> | Protein kinases                                           |
| Up   | Up           | NE                                                        |
| Up   | NE           | NE                                                        |
| Up   | Down         | NE                                                        |
| NE   | Up           | NE                                                        |
| NE   | Down         | NEK1 NEK5 NEK3 NEK4 NEK11 NEK2 group                      |
| Down | Up           | NE                                                        |
| Down | NE           | CaMKII_group, PAK_group, PDHK_group, PKD_group, SLK_group |
| Down | Down         | PKC_group                                                 |

**Table S6. Activated or inhibited protein kinase groups during starvation.** Activated ('Up') or inhibited ('Down') protein kinase groups during starvation. 'NE' indicates no enrichment.

|      |              |                                                                                                                                          |
|------|--------------|------------------------------------------------------------------------------------------------------------------------------------------|
| WT   | <i>ob/ob</i> | Signaling pathways                                                                                                                       |
| Up   | Up           | PPAR signaling pathway                                                                                                                   |
| Up   | NE           | NE                                                                                                                                       |
| Up   | Down         | NE                                                                                                                                       |
| NE   | Up           | NE                                                                                                                                       |
| NE   | Down         | AMPK signaling pathway,<br>Insulin signaling pathway                                                                                     |
| Down | Up           | NE                                                                                                                                       |
| Down | NE           | Prolactin signaling pathway,<br>RIG-I-like receptor signaling<br>pathway, Signaling pathways<br>regulating pluripotency of<br>stem cells |
| Down | Down         | NE                                                                                                                                       |

**Table S7. Activated or inhibited signaling pathways during starvation.** Activated ('Up') or inhibited ('Down') signaling pathways during starvation. 'NE' indicates no enrichment.

| Layer name            | Chemical property                    | Biological function   | Omic                                 |
|-----------------------|--------------------------------------|-----------------------|--------------------------------------|
| metabolite            | metabolite                           | metabolite            | metabolome                           |
| metabolic reaction    | -                                    | metabolic reaction    | inference                            |
| enzyme mRNA           | mRNA                                 | enzyme                | transcriptome                        |
| enzyme protein        | protein                              | enzyme                | proteome                             |
| enzyme PTM            | phosphorylation                      | enzyme                | phosphoproteome                      |
| TF mRNA               | mRNA                                 | Transcription factor  | transcriptome                        |
| TF protein            | protein                              | Transcription factor  | proteome                             |
| TF PTM                | phosphorylation                      | Transcription factor  | phosphoproteome                      |
| TF activity           | -                                    | Transcription factor  | inference                            |
| kinase mRNA           | mRNA                                 | Protein kinase        | transcriptome                        |
| kinase protein        | protein                              | Protein kinase        | proteome                             |
| kinase PTM            | phosphorylation                      | Protein kinase        | phosphoproteome                      |
| kinase activity       | -                                    | Protein kinase        | inference                            |
| signaling activity    | -                                    | signaling pathway     | inference                            |
| transporter mRNA      | mRNA                                 | transporter           | transcriptome                        |
| transporter protein   | protein                              | transporter           | proteome                             |
| transporter PTM       | phosphorylation                      | transporter           | phosphoproteome                      |
| transporter activity  | -                                    | transporter           | inference                            |
| lipid                 | lipid                                | lipid                 | lipidome                             |
| bloodmetabolite       | metabolite                           | metabolite in blood   | metabolome                           |
| FFAs and acyl species | fatty acid, acyl-CoA, acyl-carnitine | FFAs and acyl species | fatty acid, acyl-CoA, acyl-carnitine |

**Table S8: Omic layers in the global starvation-responsive metabolic network.** Classification of molecules or inferred regulators into omic layers based on biological function or chemical property.

### **Data File S1. All the analyzed data**

This file includes all the analyzed data (metabolite, lipid, FFAs and acyl species, protein, phosphorylation, transcript, plasma metabolite, and Western blotting), which were quantified in more than half of replicates in both WT and *ob/ob* mice at all the time points after food removal. Raw transcriptomic, raw proteomic, and raw phosphoproteomic data have been uploaded to public repositories (see Data and code availability section). All measured metabolomic, lipidomic, and FFAs and acyl species data are also included. Each column is a biological sample (S25-S104), whose biological conditions are summarized in the 'Sample\_information' tab. The rows are molecules or phosphorylation events. The units were nmol/mg-protein for metabolites except for glycogen, ng/μg-protein for glycogen, μM for plasma metabolites, pmol/mg-protein for lipids and FFAs and acyl species, TPM for transcripts, and arbitrary units for protein phosphorylation, and Western blotting.

### **Data File S2. Properties of the analyzed molecules**

This file includes properties of the time courses (AUC and *t*-half value) and results of statistical tests. The columns give the names of properties of the time courses (AUC and *t*-half value) and the results of statistical test (*P* values, *Q* values, and changes). The rows represent molecules. Abbreviations for WT change and *ob/ob* change columns were as follows: Up, Down, and NS represent increase, decrease and no significant changes.

### **Data File S3. KEGG Pathway Enrichment Analyses**

This file includes all the results of Fisher's exact test in KEGG Pathway Enrichment analyses for transcriptomic, proteomic, and phosphoproteomic data. The Entry, Name, Class, and Subclass are displayed for the information of each pathway. *P* values, *Q* values, and odds ratio for Fisher's exact test, and significantly changed downstream molecules are listed.

### **Data File S4. Transcription factor and protein kinase enrichment analyses**

This file includes all the results of transcription factor/kinase enrichment analysis for increased or decreased transcripts/phosphorylation events in WT and *ob/ob* mice, respectively. For transcription factors, Fisher's exact tests were conducted for each ChIP-seq experiment. The table contains experimental IDs, name of transcription factors, organ, and cell for the information of ChIP-seq experiments. *P* values, *Q* values, and odds ratio for Fisher's exact test are displayed. For protein kinases, names of kinase groups, ENSMUSG IDs of kinases, and names of kinases were displayed. *P* values, *Q* values, and odds ratio are listed.

### **Data File S5. The global starvation responsive transomic network**

This file includes all the information for nodes and edges of the global starvation-responsive transomic network. The node tab includes information of node; IDs, names, omic layers, changes in WT and *ob/ob* mice, and comparisons between them. The edge tab includes the layers and IDs of the source and target nodes, the type of regulation (Act: activation or Inh: inhibition), and the presence or absence of the regulation in WT and *ob/ob* mice (TRUE or FALSE).

### **Data File S6. The starvation responsive metabolic network**

This file includes all the information for nodes and edges of the starvation-responsive metabolic network. The node tab includes information of node: names, IDs, omic layers, significance of

responses to starvation in WT and *ob/ob* mice (TRUE or FALSE), AUC of the time courses in WT and *ob/ob* mice, *t*-half value in WT and *ob/ob* mice, and degrees and cluster IDs of the background network (network of the database) and starvation-responsive metabolic networks in WT and *ob/ob* mice. The edge tab includes the layers and IDs of the source and target nodes, the type of regulation (Act: activation or Inh: inhibition), and the presence or absence of the regulation in WT and *ob/ob* mice (TRUE or FALSE).

#### **Data File S7. The correlation metabolic network during starvation**

This file includes all the pairwise correlation coefficients among time courses of molecules in the starvation-responsive metabolic networks. Each tab represents WT or *ob/ob* mice. Both the columns and rows represent responsive molecules in the starvation-responsive metabolic network in WT and *ob/ob* mice. The pairwise correlation coefficients are displayed as a matrix in each tab.

#### **Data File S8. MSigDB Enrichment Analyses**

This file includes all the results of Fisher's exact test in MSigDB Enrichment Analyses for transcriptome and proteome data. *P* values, *Q* values, and odds ratio for Fisher's exact test, and significantly changed downstream molecules are provided for each Hallmark.

#### **Data File S9. Kinase names and ENSMUSG IDs of all the kinase groups in NetPhorest**

Kinase names, kinase groups, and ENSMUSG IDs of protein kinases are in this table.